ACADEMIC CAPSTONE & TECHNICAL EVALUATION

# Personal Developer Portfolio Website

A production-grade digital platform engineered with semantic web standards, applied RAG intelligence, and human-centered design principles.


PRESENTER

**Swetha N.**

B.E. Computer Science & Design (Class of 2028)

INSTITUTION

**Rajalakshmi Eng. College**

Chennai · CGPA: 8.65


CORE DOMAINS

**Backend & Applied AI**

RAG · Spring Boot · Django · Systems

PRODUCTION CDN

**Verified Edge Platform**

my-portfolio-...vercel.app

CORE PHILOSOPHY

# The Problem & The Purpose

Moving beyond flat resumes to establish verified engineering credibility.

THE PROBLEM

**Static PDF resumes fail to convey live engineering craftsmanship, interactive problem-solving, or system design.**

Developers need an authentic digital platform that proves capability through live code, production internship outcomes, and architectural depth.

01

### Showcase Production Engineering

Highlight genuine enterprise experience from Zoho Corporation — specifically OpenAPI RAG workflows, vector embeddings, and backend services.

02

### Prove Algorithmic Fluency

Provide verified credentials through 500+ LeetCode problems solved across Trees, Graphs, Dynamic Programming, and Competitive Hackathons.

03

### Deliver Thoughtful Ergonomics

Execute a zero-framework, accessible web experience (<50ms load time) with persistent dark/light theming and human-centered design.

USER JOURNEY & ARCHITECTURE

# Website Structure & Visitor Flow

How a technical reviewer or recruiter moves through the portfolio seamlessly.

01
**Hero**
First Impression & Role

02
**About**
Academic & Philosophy

03
**Skills**
6 Concrete Domains

04
**Projects**
7 Verified Case Studies

05
**Experience**
Zoho SWE & LeetCode

06
**Resume**
PDF & Summary Badges

07
**Contact**
Validated Form

### Editorial Dark Theme & Zero Flash

A refined midnight palette (#090D16) paired with an inline localStorage theme-engine prevents white flashes on reload while offering dark/light comfort.

### Evidence-Based Competencies

Replaced arbitrary skill percentage bars with categorized, tangible toolsets and verifiable milestone counts (500+ LeetCode problems solved).

### Frictionless Recruiter Conversion

Quick-view modals allow rapid project inspection without leaving context. One-click resume download and client-validated contact channels facilitate direct outreach.

ENGINEERING STACK

# Technology & Implementation Flow

A disciplined, zero-framework stack prioritizing speed, accessibility, and reliability.

01 · STRUCTURE

## Semantic HTML5

- ARIA landmark architecture (<header>, <main>, <nav>, <section>)
- Accessibility-first keyboard skip-link navigation
- OpenGraph & Twitter meta tags for rich social indexing

02 · STYLING

## Vanilla CSS3 System

- CSS Custom Properties (:root) for instant theme variables
- Modern CSS Grid & Flexbox for multi-device responsiveness
- Subtle glassmorphic backdrops and fluid clamp() typography

03 · INTERACTIVITY

## Vanilla JavaScript (ES6+)

- Zero external framework bloat — pure browser native code
- Theme persistence controller with localStorage synchronization
- Active scroll-spy navigation listener & Quick View modal logic

04 · INFRASTRUCTURE

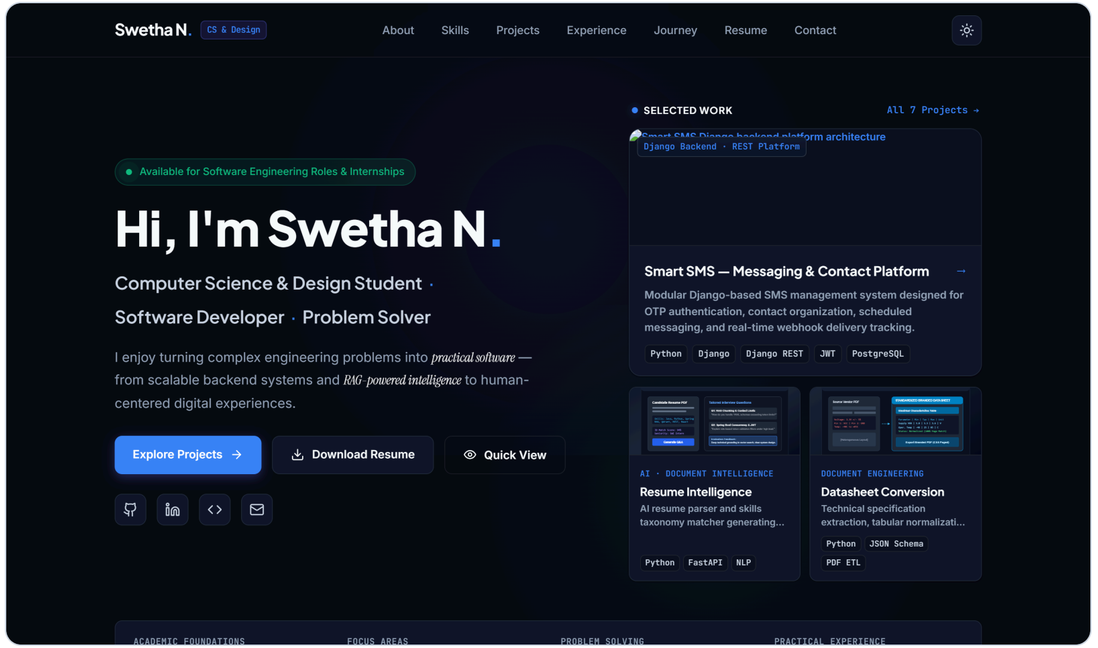
## Git, GitHub & Vercel

- Production version control on GitHub main branch
- Automated continuous deployment via Vercel Edge CDN
- Zero build-step pipeline ensuring instantaneous deployments

VISUAL PROOF

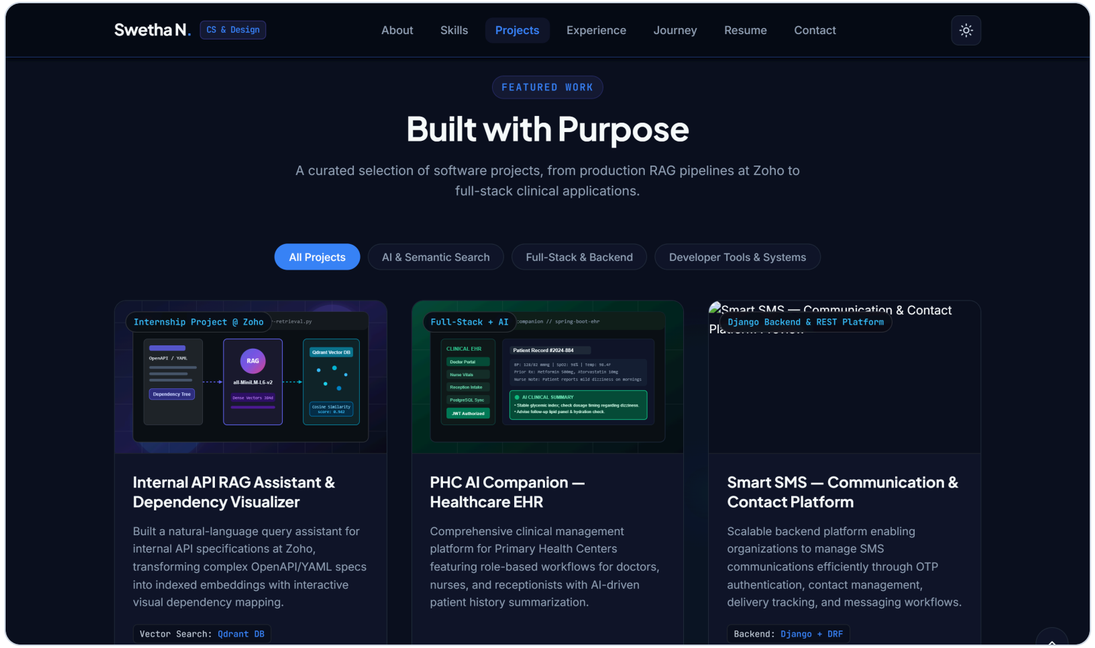
# The Implemented Website

Genuine high-resolution captures showing the live design system in action.




## Hero & Selected Work Overview

Balanced personal introduction, role pill, live status indicator, and multi-project quick cards with direct action buttons.




## Featured Projects Showcase

Deep engineering breakdowns: Zoho OpenAPI RAG Assistant, Healthcare EHR Companion, and Smart SMS Django Messaging Platform.

PROJECT SUMMARY

# Outcomes & Future Roadmap

Verified deliverables, technical evaluation, and future engineering trajectory.

VERIFIED PROJECT OUTCOMES

**✓ Production-Ready Deployment**

Successfully hosted on Vercel Edge with zero downtime, instant SSL, and worldwide CDN distribution.

**✓ Zero-Dependency Architecture**

Pure HTML5, CSS3, and ES6+ delivering sub-50ms render speed without heavy client bundle overhead.

**✓ Dual-Theme Precision**

Flawless dark and light modes with localStorage persistence and zero layout shift.

**✓ Authentic Technical Depth**

Seamlessly integrates real Zoho Corporation RAG internship contributions and 500+ LeetCode milestones.

FUTURE ENGINEERING ROADMAP

**→ Live LeetCode & GitHub Sync**

Dynamic automated pipeline refreshing solved DSA milestones and live repo activity.

**→ Interactive Semantic RAG Playground**

In-browser live demonstration of OpenAPI and YAML semantic vector retrieval with LangChain.

**→ Automated Lighthouse CI Gates**

Continuous integration checks enforcing 98+ accessibility, performance, and SEO scores on commits.

**→ Ambient WebGL Micro-Interactions**

Subtle GPU-accelerated interactive canvas shaders for featured project case studies.

CAPSTONE EVALUATION & CONCLUSION

# Thank You!

Open for Technical Questions, Architectural Discussions & Feedback


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LIVE DEPLOYMENT

**Vercel Edge Platform**

my-portfolio-...vercel.app

GITHUB REPOSITORY

**swethanagaraj2007**

github.com/swethanagaraj2007

PROFESSIONAL CONNECT

**LinkedIn Network**

linkedin.com/in/swetha-n-005697334